

Crystal-field effects on the transport properties of *f*-electron systems

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ABSTRACT:

In this study, we examine the crystal-field effects on the single-ion states or various heavy fermion elements and systems. Using Stevens parameters and exact diagonalization, we generate a first-order approximation of crystal-field disorder as systems undergo phase transitions between D2, D3, and D4 symmetries. Furthermore, we examine the effect on transport properties, as determined from the state, and apply it to the known cerium and ytterbium systems.

METHODOLOGY:

Treating surrounding ions in a molecule as point charges we can approximate the electrical potential acting on a magnetic ion. The potential can be expanded in the terms of spherical harmonics.

$$V_{CF}(r_i) = \sum_{l,m} A_{ml} r_l Y_{lm}(r_i)$$

For rare earth ions, we consider the total angular momentum quantum number J and the secondary quantum number J_z due to the strong spin-orbit interactions that exists (leading to a mixing between the spin and orbital angular momentum states). The crystal field Hamiltonian matrix thus represents the energy splitting of J_z due to the inferred crystal field of Ce^{3+} . The Hamiltonian is constructed in a similar manner to the potential but is now created by summing tensor operators that act on the J_z basis states.

Results

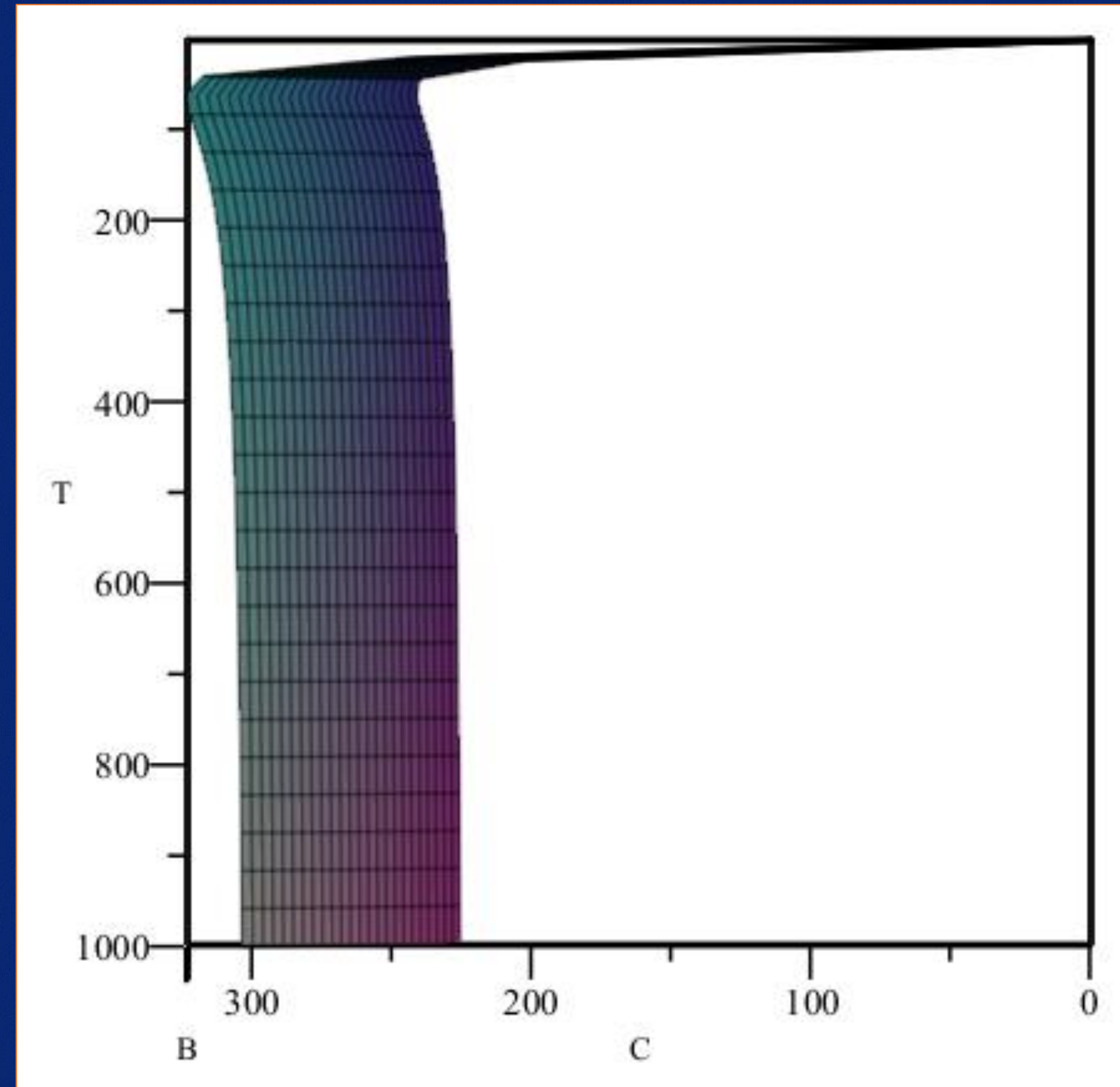


Fig 1. Heat capacity $C(T, B, A)$ for CeI_3 for $a = 0$ (i.e- corresponding to the D3 symmetry).

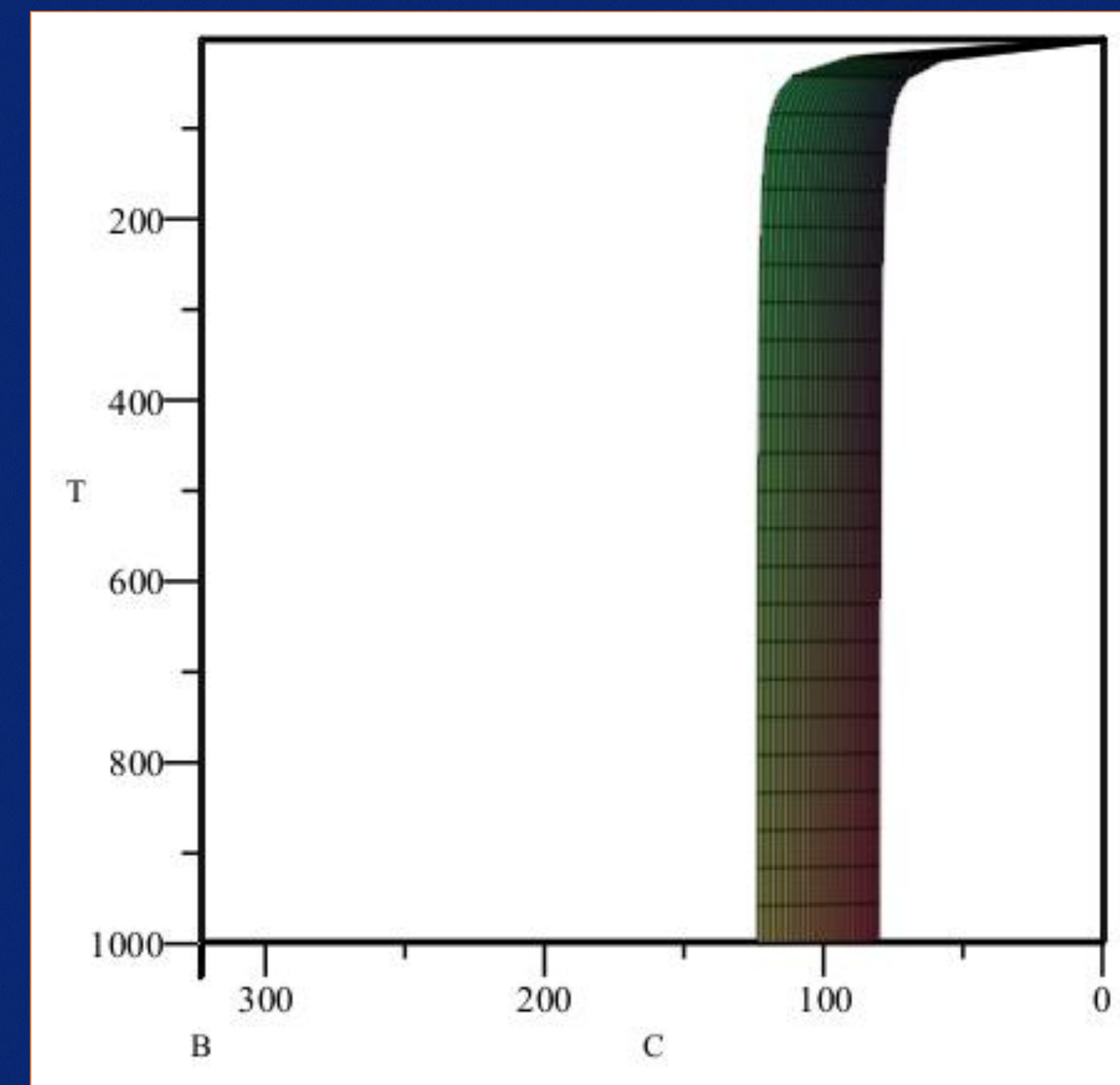


Fig 2. Heat capacity $C(T, B, A)$ for CeI_3 for $a = 1$ (i.e- corresponding to the D4 symmetry).

$$\mathcal{H} = \begin{pmatrix} B & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & B \end{pmatrix}$$

Equation 1. Representation of the block diagonalized crystal field Hamiltonian

$$B_{d_3} = \begin{bmatrix} 10B_{20} + 60B_{40} & 0 & 3\sqrt{10} B_{43} \\ 0 & -8B_{20} + 120B_{40} & 0 \\ 3\sqrt{10} B_{43} & 0 & -8B_{20} + 120B_{40} \end{bmatrix}$$

Equation 3. Crystal field Hamiltonian for trigonal D_3 symmetry

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symmetry	point group	B_2^0	$B_2^{\pm 1}$	$B_2^{\pm 2}$	B_4^0	$B_4^{\pm 1}$	$B_4^{\pm 2}$	$B_4^{\pm 3}$	$B_4^{\pm 4}$	B_6^0	$B_6^{\pm 1}$	$B_6^{\pm 2}$	$B_6^{\pm 3}$	$B_6^{\pm 4}$
triclinic	C_i, C_1	+	±	±	+	±	±	±	±	+	±	±	±	±
monoclinic	C_2, C_s, C_{2h}	+	±		+	±				+	±			
rhombic	C_{2v}, D_2, D_{2h}	+		+	+		+			+		+		+
tetragonal	C_4, S_4, C_{4h}	+			+			±	+	+			±	+
tetragonal	$D_4, C_{4v}, D_{2d}, D_{4h}$	+			+			±	+	+			±	+
trigonal	C_3, S_6	+			+					+			±	
trigonal	D_3, C_{3v}, D_{3d}	+			+			±	+	+			±	+
hexagonal	C_6, C_{3h}, C_{6h}	+			+					+				
hexagonal	$D_6, C_{6v}, D_{3h}, D_{6h}$	+			+					+				
cubic	T, T_h	+			+					+				
cubic	T_d, O, O_h	+			+					+				

TABLE I. Allowed Stevens parameters (B_q^k) for different crystal symmetries and point groups. A '+' means the parameter is allowed; '±' means its real or imaginary part may be nonzero.

For an ion in thermal equilibrium, the partition function is

$$Z(T, B, a) = \sum_{i=1}^6 \exp[-\beta(E_i(a) - Bm_i)]$$

where $\beta = \frac{1}{k_b T}$ and the free energy F is

$$F = -\frac{1}{\beta} \ln Z.$$

The constant field heat capacity C is obtained from the temperature derivative of the internal energy,

$$C = \frac{\partial}{\partial T} \left(\frac{\partial}{\partial \beta} \ln Z \right),$$

which is equivalent to the second derivative of $\ln Z$ with respect to β :

$$C = \frac{\partial^2}{\partial \beta^2} \ln Z |_{\beta=1/T}.$$

This is exactly what we compute symbolically/numerically in Maple.

We thus have a general method for producing the crystal field Hamiltonian. To analytically measure the specific heat capacity of our compound, we estimate the Stevens parameters using a point charge model. We utilized the python library PyCrystalField which uses the ligand environment which is found from the .cif file of a compound. We look at CeI_3 . Using Maple, we find the eigenvalues and eigenvectors for each symmetry (D_2, D_3, D_4) and the total Hamiltonian H is first diagonalized to obtain the crystal-field-split energy levels, We interpolate between symmetries using a mixing parameter a

$$H(a) = a H_2 + (1 - a) H_4 \text{ (example)}$$